

Entity with attributes

User :

Column Name	Data Type	Constraints
user_id	VARCHAR(50)	Primary Key
username	VARCHAR(100)	Not Null
email	VARCHAR(100)	Unique, Not Null
Role (keep the below entity and make it as foreign key)	VARCHAR(50)	Values: EMPLOYEE / MANAGER / HR / FINANCE / ADMIN / SUPERADMIN
created_at	TIMESTAMP	Default: Current Date & Time

(this are extension only for the understanding)

Employee:

Column Name	Data Type	Constraints
employee_id	VARCHAR(50)	Primary Key
user_id	VARCHAR(50)	Foreign Key → User(user_id)
designation	VARCHAR(100)	
department	VARCHAR(100)	
salary_grade	DOUBLE(2)	

Manager :

Column Name	Data Type	Constraints
employee_id	VARCHAR(50)	Primary Key
user_id	VARCHAR(50)	Foreign Key → User(user_id)
designation	VARCHAR(100)	

HR :

Column Name	Data Type	Constraints
hr_id	VARCHAR(50)	Primary Key
user_id	VARCHAR(50)	Foreign Key → User(user_id)
department	VARCHAR(50)	

Policy :

Column Name	Data Type	Constraints
policy_id	VARCHAR(50)	Primary Key
designation	VARCHAR(100)	
travel_class	VARCHAR(50)	
max_limit	DECIMAL(12,2)	
rules	TEXT	

TravelRequest :

Column Name	Data Type	Constraints
request_id	VARCHAR(50)	Primary Key
employee_id	VARCHAR(50)	Foreign Key → Employee(employee_id)
purpose	VARCHAR(255)	
location	VARCHAR(56)	
from_date	DATE	
to_date	DATE	
travel_mode	VARCHAR(50)	Values: Air / Train / Bus / Cab / Personal Vehicle
estimated_cost	DECIMAL(12,2)	
food_estimated_cost	DECIMAL(12,2)	
accommodation_estimated_cost	DECIMAL(12,2)	
status	VARCHAR(50)	Values: PENDING / APPROVED / REJECTED
created_at	TIMESTAMP	Default: Current Date & Time
policy_id	VARCHAR(50)	Foreign Key → Policy(policy_id)
supporting_document	VARCHAR(255)	File path / URL
approver_id	VARCHAR(50)	Foreign Key → Manager(manager_id) OR User(user_id)

Approval :

Column Name	Data Type	Constraints
approval_id	VARCHAR(50)	Primary Key
request_id	VARCHAR(50)	Foreign Key → TravelRequest(request_id)
approver_id	VARCHAR(50)	Can refer to Manager(manager_id) or HR(hr_id)
approver_type	VARCHAR(50)	Values: MANAGER / HR
status	VARCHAR(50)	Values: APPROVED / REJECTED
remarks	VARCHAR(255)	
approved_at	TIMESTAMP	

Payment :

Column Name	Data Type	Constraints
payment_id	VARCHAR(50)	Primary Key
request_id	VARCHAR(50)	Foreign Key → TravelRequest(request_id)
amount	DECIMAL(12,2)	
payment_mode	VARCHAR(50)	Values: BANK / CARD / WALLET

status	VARCHAR(50)	Values: PENDING / PROCESSED
processed_at	TIMESTAMP	

EmployeeLevelAssignment :

Column Name	Data Type	Constraints
assignment_id	VARCHAR(50)	Primary Key
user_id	VARCHAR(50)	Foreign Key → User(user_id)
level	VARCHAR(50)	Values: L1, L2, L3, L4
effective_from	DATE	
effective_to	DATE	Nullable

ReportingLine :

Column Name	Data Type	Constraints
report_id	VARCHAR(50)	Primary Key
employee_id	VARCHAR(50)	Foreign Key → Employee(employee_id)
manager_id	VARCHAR(50)	Foreign Key → Manager(manager_id)
effective_from	DATE	
effective_to	DATE	Nullable

Attachment :

Column Name	Data Type	Constraints
attachment_id	VARCHAR(50)	Primary Key
request_id	VARCHAR(50)	Foreign Key → TravelRequest(request_id)
file_url	VARCHAR(255)	
file_type	VARCHAR(50)	PDF, JPG, PNG, DOC, etc.
uploaded_by	VARCHAR(50)	Foreign Key → User(user_id)
uploaded_at	TIMESTAMP	

TicketStatusHistory :

Column Name	Data Type	Constraints
history_id	VARCHAR(50)	Primary Key
request_id	VARCHAR(50)	Foreign Key → TravelRequest(request_id)
old_status	VARCHAR(50)	
new_status	VARCHAR(50)	
changed_by	VARCHAR(50)	Foreign Key → User(user_id)
changed_at	TIMESTAMP	

Role :

Column Name	Data Type	Constraints
role_id	VARCHAR(50)	Primary Key
role_name	VARCHAR(50)	Unique
description	VARCHAR(255)	

UserRole :

Column Name	Data Type	Constraints
user_role_id	VARCHAR(50)	Primary Key
user_id	VARCHAR(50)	Foreign Key → User(user_id)
role_id	VARCHAR(50)	Foreign Key → Role(role_id)

RolePermission:

Column Name	Data Type	Constraints
user_role_id	VARCHAR(50)	Primary Key
user_id	VARCHAR(50)	Foreign Key → User(user_id)
role_id	VARCHAR(50)	Foreign Key → Role(role_id)

Delegation:

Column Name	Data Type	Constraints
delegation_id	VARCHAR(50)	Primary Key
from_user_id	VARCHAR(50)	Foreign Key → User(user_id)
to_user_id	VARCHAR(50)	Foreign Key → User(user_id)
start_date	DATE	
end_date	DATE	

Notification :

Column Name	Data Type	Constraints
notification_id	VARCHAR(50)	Primary Key
request_id	VARCHAR(50)	Foreign Key → TravelRequest(request_id)
recipient_id	VARCHAR(50)	Foreign Key → User(user_id)
type	VARCHAR(50)	Email / In-App / SMS
message	VARCHAR(255)	
created_at	TIMESTAMP	
sent_at	TIMESTAMP	

Query table:

Query	Solution:
1. Need of the implementation of the delegation (if the responsible person is on leave)	Query 21(r)
2. Need of the implementation of the urgent query template ?	Query 8,22
3. Need of the post implementation query template ?	Query 26
4. Read existing travel policy document for reference .	
5. Redundant entity : Approval	
6. Think of two side roles	
7. Addition of the travel class (entity) in relation with travel req and user	
8. Change the name ofn the entity userlevelassign	
9. Also add the class and relation between them	

